

SUBREGULAR REDUCTION AND BINARY COLLISIONS

1.1 AN EXPLICIT SUBREGULAR REDUCTION OF $\mathfrak{sl}_3$

The subregular reduction has a nonzero weight-one current and two even charged fields. Their vacuum products determine both the reduction map and its normalization. An integral good grading gives a complex with two charged ghost pairs and no neutral ghosts.

Work first over $R = \mathbb{C}[k]$. All tensor products and direct sums are algebraic. The symbol k is an indeterminate of cohomological degree zero. For matrix units E_{ij} , put

$$a = E_{12}, \quad e = E_{13}, \quad A = E_{21}, \quad B = E_{31}, \quad u = E_{23}, \quad v = E_{32},$$

$$h = \text{diag}(1, -1, 0), \quad j = \frac{1}{3} \text{diag}(1, 1, -2), \quad x = \frac{1}{3} \text{diag}(2, -1, -1), \quad f = -E_{21}.$$

Use the invariant form $(s, t) = \text{tr}(st)$ on $\mathfrak{sl}_3$. The positive eigenspace of $\text{ad } x$ is the abelian algebra $\mathfrak{m} = Ra \oplus Re$, in degree one. The degree-zero space is spanned by h, j, u, v , and the degree-minus-one space by A, B . Direct matrix commutation gives

$$[x, f] = -f, \quad \mathfrak{sl}_3^f = Rj \oplus Ru \oplus RA \oplus RB.$$

Thus every eigenvalue on the centralizer is nonpositive. The character $(f, -)|_{\mathfrak{m}}$ takes values $-1, 0$ on a, e . This is the good-grading convention for quantum reduction in [1, §1, pp. 3–6]. Changing the sign of f does not change its nilpotent orbit.

Let $V_R(\mathfrak{sl}_3)$ be the universal affine vacuum vertex algebra with

$$[s_\lambda t] = [s, t] + k \text{tr}(st) \lambda \mathbf{1}.$$

Tensor it with charged free fermions b_1, b_2, c_1, c_2 , whose only nonzero brackets are

$$[b_i c_j] = [c_j b_i] = \delta_{ij} \mathbf{1}.$$

Affine generators have cohomological degree zero, while $|b_i| = -1$ and $|c_i| = 1$. Parity is cohomological degree modulo two. Write C_R for this differential vertex superalgebra's underlying vertex superalgebra. Use the mode convention

$$Y(s, z) = \sum_{n \in \mathbb{Z}} s_{(n)} z^{-n-1}, \quad [s_\lambda t] = \sum_{n \geq 0} \frac{\lambda^n}{n!} s_{(n)} t, \quad : st := s_{(-1)} t.$$

All generating modes with $n \geq 0$ annihilate the vacuum. Every state is a finite sum of ordered negative-mode monomials.

Define the odd field and its zero mode by

$$q =: (a - \mathbf{1})c_1 : + : e c_2 :, \quad D = q_{(0)}.$$

The residue of $z^{-1}dz$ is one. There are no singular contractions in $q(z)q(w)$, so $D^2 = 0$. The zero-mode commutator formula makes D a derivation of every vertex product and gives $D\partial = \partial D$. In particular,

$$\begin{aligned} Db_1 &= a - \mathbf{1}, & Db_2 &= e, & Dc_i &= Da = De = 0, \\ Dh &= -2 : ac_1 : - : ec_2 :, & Dj &= - : ec_2 :, & Du &= : ec_1 :, \\ Dv &= : ac_2 :, & DA &= : hc_1 : - : uc_2 : + k\partial c_1, \\ DB &= - : vc_1 : + \frac{1}{2} : (h + 3j)c_2 : + k\partial c_2. \end{aligned}$$

I.I.I Four closed fields

Set $N_i = b_i c_i$ and define

$$H = h + 2N_1 + N_2, \quad J = j + N_2, \quad U = u - : b_2 c_1 :, \quad X = v - : b_1 c_2 :. \quad (1.1)$$

Let C_R^- be the vertex subalgebra generated by $H, J, U, X, A, B, c_1, c_2$. Its cohomological degrees are nonnegative.

LEMMA I.I.I. The subalgebra C_R^- is preserved by D . Its differential satisfies

$$\begin{aligned} DH &= -2c_1, & DX &= c_2, & DJ &= DU = Dc_i = 0, \\ DA &= : Hc_1 : - : Uc_2 : + (k+3)\partial c_1, \\ DB &= - : Xc_1 : + \frac{1}{2} : (H + 3J)c_2 : + (k+3)\partial c_2. \end{aligned} \quad (1.2)$$

The following four even states are closed:

$$\begin{aligned} J, \quad U, \quad P &= A + \frac{1}{4} : HH : + : UX : + \frac{k+2}{2} \partial H, \\ V &= B - \frac{1}{2} : HX : - \frac{3}{2} : JX : - (k+2)\partial X. \end{aligned} \quad (1.3)$$

Proof. The affine brackets and the ghost brackets show that C_R^- is a vertex subalgebra with a PBW basis in its displayed generators. For example,

$$[H_\lambda H] = (2k+5)\lambda \mathbf{1}, \quad [H_\lambda J] = \lambda \mathbf{1}, \quad [U_\lambda X] = -\frac{1}{2}H + \frac{3}{2}J + (k+1)\lambda \mathbf{1}.$$

The bracket of A or B with a modified current remains a linear combination of A, B . The brackets of modified degree-zero currents close among themselves and scalar terms. Their brackets with c_i are linear combinations of the c_i .

Normal ordering gives the identities

$$\begin{aligned} : N_i c_i : &:= -\partial c_i, & : N_2 c_1 : + : (: b_2 c_1 :) c_2 : &:= -\partial c_1, \\ : (: b_1 c_2 :) c_1 : + : N_1 c_2 : &:= -\partial c_2. \end{aligned}$$

They follow by moving the rightmost ghost mode through the two modes of the normal product. Substitution in DA, DB proves (1.2). The quasi-commutativity formula gives

$$: c_1 H : - : H c_1 : = 2\partial c_1, \quad : c_1 X : - : X c_1 : = -\partial c_2.$$

Therefore

$$\begin{aligned} D(\frac{1}{4} : HH :) &= - : H c_1 : - \partial c_1, & D(: UX :) &= : U c_2 :, \\ D(-\frac{1}{2} : HX :) &= : c_1 X : - \frac{1}{2} : H c_2 :, & D(-\frac{3}{2} : JX :) &= -\frac{3}{2} : J c_2 :. \end{aligned}$$

The remaining derivative terms in DP, DV cancel exactly. $\square$

Let W_R denote the vertex subalgebra of C_R^- generated by J, U, P, V , with zero differential.

THEOREM I.1.2 (*Subregular reduction map*). The inclusion

$$\iota : (W_R, \mathfrak{o}) \hookrightarrow (C_R, D)$$

is a strict unital quasi-isomorphism of differential vertex superalgebras over R . Ordered normally ordered monomials in all derivatives of J, U, P, V form an R -basis of W_R . For every $k_o \in \mathbb{C}$, specialization gives

$$H^r(C_R \otimes_R \mathbb{C}_{k_o}, D) = \begin{cases} W_R \otimes_R \mathbb{C}_{k_o}, & r = \mathfrak{o}, \\ \mathfrak{o}, & r \neq \mathfrak{o}. \end{cases}$$

Here $\mathbb{C}_{k_o} = R/(k - k_o)$, including $k_o = -3$. The degree-zero vertex algebra is the subregular quantum reduction for $(\mathfrak{sl}_3, x, f)$.

Proof. Let C_R^+ be generated by $a - \mathbf{1}, e, b_1, b_2$. Its vertex products are commutative and its differential is preserved. Normally ordered multiplication gives a linear map of complexes

$$C_R^- \otimes_R C_R^+ \longrightarrow C_R, \quad s \otimes t \longmapsto s_{(-1)}t.$$

The tensor differential is $D(s \otimes t) = Ds \otimes t + (-1)^{|s|} s \otimes Dt$. To prove bijectivity, assign filtration degrees zero to derivatives of a, e, b_i , one to derivatives of h, j, u, v, c_i , and two to derivatives of A, B . All singular brackets lower total filtration degree by at least one. The associated graded is the polynomial algebra in the even jets tensored with the exterior algebra in the odd jets. The changes $a \mapsto a - \mathbf{1}$ and (1.1), including all derivatives, are polynomial and invertible there. Thus multiplication induces a bijection on associated graded modules. Induction on the nonnegative filtration degree proves the stated linear bijection. The filtration need not have finite-dimensional pieces. Every individual state has finite degree. This is a tensor decomposition of complexes, not of vertex algebras.

Write $a'_r = \partial^r(a - \mathbf{1})$, $e_r = \partial^r e$, and $b_{ir} = \partial^r b_i$. The complex C_R^+ has differential

$$D_+ = \sum_{r \geq \mathfrak{o}} \left(a'_r \frac{\partial}{\partial b_{1r}} + e_r \frac{\partial}{\partial b_{2r}} \right).$$

Odd derivatives act from the left. For $s_+ = \sum_r (b_{1r} \partial_{a'_r} + b_{2r} \partial_{e_r})$, one has $D_+ s_+ + s_+ D_+ = N_+$, where N_+ counts these factors. Division by each positive integer eigenvalue gives a contraction onto $R\mathbf{1}$. All sums are finite on a state. Consequently $C_R^- \hookrightarrow C_R$ is a quasi-isomorphism.

On C_R^- assign degree one to derivatives of H, J, U, X, c_i , and degree two to derivatives of A, B . The associated graded differential is

$$dH = -2c_1, \quad dX = c_2, \quad dJ = dU = dc_i = \mathfrak{o},$$

$$dA = Hc_1 - Uc_2, \quad dB = -Xc_1 + \frac{1}{2}(H + 3J)c_2.$$

It commutes with all formal derivatives. Introduce the polynomial variables

$$p = A + H^2/4 + UX, \quad v_o = B - HX/2 - 3JX/2.$$

This triangular change is invertible, and $dp = dv_o = \mathfrak{o}$. Hence the graded complex is

$$R[\partial^r J, \partial^r U, \partial^r p, \partial^r v_o : r \geq \mathfrak{o}] \otimes_R (R[H_r, X_r] \otimes \Lambda(c_{1r}, c_{2r}), d).$$

On the second factor,

$$s_- = -\frac{1}{2} \sum_r H_r \frac{\partial}{\partial c_{1r}} + \sum_r X_r \frac{\partial}{\partial c_{2r}}$$

satisfies $ds_- + s_- d = N_-$. Thus the graded cohomology is precisely the first polynomial factor in degree zero.

The explicit cycles (1.3) lift its four generators. Their ordered monomials have independent leading symbols, which proves linear independence. Given a degree-zero closed state, subtract a linear combination of these monomials with the same leading symbol. The filtration degree decreases. Repeating terminates and proves that these monomials span $\ker(D|_{(C_R^-)^\circ})$. There are no degree-minus-one states in C_R^- . Thus this kernel is W_R itself. For a closed state in positive cohomological degree, the graded contraction produces a preimage of its leading symbol. Subtracting its differential lowers the filtration degree and again terminates. It follows that $H^{r>0}(C_R^-) = 0$.

These arguments use only division by nonzero integers and the constant 2. They therefore apply over R and after every specialization. No inverse of $k+3$ enters this proof. This proves all assertions, including base change and the identification with the defining quantum reduction. $\square$

Theorem 1.1.2 gives an explicit instance of the structure theorem in [1, Theorem 4.1 and its proof, pp. 10–12]. The displayed contraction proves the required assertions on this universal vacuum carrier directly. Write $C_k = C_R \otimes_R \mathbb{C}_k$ and $W_k = W_R \otimes_R \mathbb{C}_k$ for a fixed complex level.

1.1.2 The Bershadsky–Polyakov products

For $k \neq -3$, set $t = k+3$ and

$$\begin{aligned} G^+ &= U, & G^- &= -V, & T &= \frac{P + \frac{3}{4} : JJ : - \frac{3}{2} \partial J}{t}, \\ \kappa_J &= \frac{2k+3}{3}, & c &= -\frac{(2k+3)(3k+1)}{k+3}. \end{aligned} \tag{1.4}$$

The sign in $G^- = -V$ fixes the cubic vacuum term. The fields $G^\pm$ are even.

PROPOSITION 1.1.3. The following brackets hold as identities of states in C_k , not only in cohomology:

$$\begin{aligned} [J_\lambda J] &= \kappa_J \lambda \mathbf{1}, & [J_\lambda G^\pm] &= \pm G^\pm, & [G^\pm_\lambda G^\pm] &= 0, \\ [T_\lambda J] &= (\partial + \lambda)J, & [T_\lambda G^\pm] &= (\partial + \frac{3}{2}\lambda)G^\pm, \\ [T_\lambda T] &= (\partial + 2\lambda)T + \frac{c}{12} \lambda^3 \mathbf{1}. \end{aligned} \tag{1.5}$$

$$\begin{aligned} [G^+_\lambda G^-] &= 3 : JJ : + \frac{3(k+1)}{2} \partial J - tT + 3(k+1)\lambda J \\ &\quad + \frac{(k+1)(2k+3)}{2} \lambda^2 \mathbf{1}. \end{aligned} \tag{1.6}$$

Consequently W_k is the universal Bershadsky–Polyakov vertex algebra in this normalization.

Proof. Use the noncommutative Wick formula

$$[s_\lambda : ab :] = : [s_\lambda a] b : + (-1)^{p(s)p(a)} : a [s_\lambda b] : + \int_0^\lambda [[s_\lambda a]_\mu b] d\mu$$

and quasi-commutativity : $ab : -(-1)^{p(a)p(b)} : ba := \int_{-\partial}^{\circ} [a_{\lambda} b] d\lambda$. For $M_{ij} =: b_i c_j$, their input ghost identities are

$$\begin{aligned} [M_{ij\lambda} M_{lm}] &= \delta_{jl} M_{im} - \delta_{im} M_{lj} + \delta_{jl} \delta_{im} \lambda \mathbf{1}, \\ [M_{ij\lambda} b_l] &= \delta_{jl} b_i, \quad [M_{ij\lambda} c_l] = -\delta_{il} c_j. \end{aligned}$$

Affine-ghost mixed brackets vanish. These formulas immediately give $[J_{\lambda} J] = \kappa_J \lambda$ and $[J_{\lambda} U] = U$. The next substitutions give

$$\begin{aligned} [J_{\lambda} V] &= -V, \\ [J_{\lambda} P] &= \tfrac{3}{2} \lambda J + \tfrac{2k+3}{2} \lambda^2 \mathbf{1}, \\ [U_{\lambda} V] &= P - \tfrac{2}{4} : JJ : - \tfrac{3(k+2)}{2} \partial J - 3(k+1) \lambda J - \tfrac{(k+1)(2k+3)}{2} \lambda^2 \mathbf{1}. \end{aligned}$$

For the last identity, insert $V = B - : HX : / 2 - 3 : JX : / 2 - (k+2) \partial X$. The required brackets are

$$[U_{\lambda} B] = A, \quad [U_{\lambda} H] = U, \quad [U_{\lambda} J] = -U, \quad [U_{\lambda} X] = -H/2 + 3J/2 + (k+1)\lambda.$$

The two Wick integrals combine to $\lambda U_{(0)} X + \frac{k+1}{2} \lambda^2 \mathbf{1}$. Expanding the four displayed summands, and using quasi-commutativity for H, J , yields the asserted formula. Substitution of (1.4) proves (1.6). The formula for $[J_{\lambda} P]$ and $[J_{\lambda} : JJ :] = 2\kappa_J \lambda J$ give the stated T, J bracket.

For the conformal normalization, let S be the affine Sugawara field and put

$$L^{\text{full}} = S + \partial x + \sum_{i=1}^2 : (\partial b_i) c_i :.$$

Its affine generator weights are $1 - \deg_x(s)$. Each ghost pair has weights $(0, 1)$ and central charge -2 . The field q is primary of weight one, so $DL^{\text{full}} = 0$. The Sugawara and ghost contractions give

$$c_L = \frac{8k}{t} - 8k - 4, \quad [L^{\text{full}}_{\lambda} J] = (\partial + \lambda) J - \frac{\kappa_J}{2} \lambda^2 \mathbf{1}.$$

Here $(x, x) = 2/3$, $(x, j) = 1/3$, and the second ghost pair contributes $-\lambda^2/2$. There is an explicit homotopy identity:

$$t(L^{\text{full}} - T - \tfrac{1}{2} \partial J) = D(: b_1 A : + : b_2 B :). \quad (1.7)$$

For its verification, write the affine Sugawara numerator as

$$tS = \tfrac{1}{4} : hh : + \tfrac{3}{4} : jj : + \tfrac{1}{2} (: aA : + : Aa : + : eB : + : Be : + : uv : + : vu :).$$

Use $Aa := aA - \partial h$, $Be := eB - \partial(h + 3j)/2$, and $vu := uv + \partial(h - 3j)/2$. Substitute (1.1)–(1.3). The remaining terms are $(a-1)A : + : eB : - : b_1 DA : - : b_2 DB :$, which equal the right side of (1.7) by the odd derivation rule.

Improving L^{full} by $-\partial J/2$ changes the central charge by $3\kappa_J$. It makes J primary and gives $G^{\pm}$ weights $3/2$. Here charge is the eigenvalue of $J_{(0)}$. Under $L^{\text{full}}_{(1)}$, the states H, J, U, X have weight one and A, B, P, V have weight two. The improvement adds half the charge to these weights. Its cohomology class is $[T]$. The inclusion $\mathcal{W}_k \rightarrow C_k$ is injective on cohomology. Thus its conformal identities hold strictly in $\mathcal{W}_k$. The value $c = c_L + 3\kappa_J$ is exactly (1.4). Higher products $T_{(n)} G^{\pm}$, $n \geq 2$, have charge ± 1 and weight at most $1/2$. The PBW basis has no such states. Products $G^{\pm}_{(n)} G^{\pm}$, $n \geq 0$, have charge ± 2 and weight at most two. The smallest PBW weight with either charge is three. These products therefore vanish. This proves all brackets. They agree with the universal presentation in [2, §2, p. 2]. The PBW theorem proves that this presentation introduces no further relations. $\square$

COROLLARY I.1.4 (*Special levels and reflection*). The current vacuum coefficient is $\kappa_J = (2k + 3)/3$. The G^+ , G^- cubic coefficient is $(k + 1)(2k + 3)$, and the T , T quartic coefficient is $c/2$. All three vanish simultaneously at $k = -3/2$. At $k = -1/2$ they are $2/3, 1, 1/5$, respectively. For $k' = -k - 6$ and $k \neq -3$, the reflected conformal charges obey

$$c(k) + c(k') = 50.$$

Equality $c(k) = c(k')$ holds precisely when $(k + 3)^2 = -4$. Hence a conformal vertex isomorphism at reflected levels is impossible for all other noncritical parameters.

Proof. The common root follows from the first coefficient and direct substitution in the other two. At $k = -1/2$, the charge is $2/5$. Writing $t = k + 3$ gives $c(k) = 25 - 6t - 24/t$ and $c(k') = 25 + 6t + 24/t$. Their equality is equivalent to $t^2 = -4$. A conformal unit map preserves $T_{(3)}T = (c/2)\mathbf{1}$. At the two exceptional parameters, this calculation states only equality of charges. It does not construct an isomorphism. $\square$

I.1.3 Unital binary collisions

Fix a complex level k , including $k = -3$, and an affine coordinate z on $\mathbb{A}^1$. The map $\iota : W_k \rightarrow C_k$ commutes with translation and every vertex product. For a differential vertex algebra F , its right differential-operator module is $\mathcal{A}_F = \omega_{\mathbb{A}^1} \otimes_{\mathbb{C}} F$, with

$$(g(z) dz \otimes a) \cdot \partial_z = -g'(z) dz \otimes a - g(z) dz \otimes \partial a.$$

In left-module notation, put

$$\mathcal{P}_F = \mathbb{C}[z, w, (z - w)^{-1}] \otimes F \otimes F, \quad \mathcal{R}_F = F \otimes \mathbb{C}[w] \otimes \xi^{-1} \mathbb{C}[\xi^{-1}], \quad \xi = z - w.$$

The source connections are $\partial_z^{\text{base}} + \partial_F \otimes \mathbf{1}$ and $\partial_w^{\text{base}} + \mathbf{1} \otimes \partial_F$. The target actions are

$$\partial_z = \partial_\xi, \quad \partial_w = \partial_w^{\text{base}} - \partial_\xi + \partial_F.$$

Multiplication by ξ on the target is taken modulo regular powers. Tensoring with $dz \wedge dw$ gives the corresponding right modules. The orientation is $dz \wedge dw = d\xi \wedge dw$, with $\text{Res } \xi^{-1} d\xi = \mathbf{1}$. The tensor differential on $\mathcal{P}_F$ uses the sign $(-1)^{|a|}$ on its second state. Define the full meromorphic collision by

$$\mu_F(g(z, w) \otimes a \otimes b) = \text{pp}_{\xi=0}(g(w + \xi, w)Y(a, \xi)b).$$

Here pp retains the strictly negative powers of ξ . The answer is finite by vertex truncation and the finite pole order of g . Both μ_F and ι commute with the internal differential.

PROPOSITION I.1.5. The map $\mathcal{A}_{W_k} \rightarrow \mathcal{A}_{C_k}, g dz \otimes a \mapsto g dz \otimes \iota(a)$, is a unital quasi-isomorphism of differential chiral algebras on this affine chart. Its tensorwise maps make the square of μ_{W_k} and μ_{C_k} commute exactly. They are quasi-isomorphisms on $\mathcal{P}$ and $\mathcal{R}$, and on their collision mapping cones. In particular, the vacuum terms are preserved:

$$\mu_{C_k}(J \otimes J) = \kappa_J \xi^{-2} \mathbf{1},$$

and, for $k \neq -3$,

$$\begin{aligned} \mu_{C_k}(G^+ \otimes G^-) &= (k + 1)(2k + 3) \xi^{-3} \mathbf{1} + 3(k + 1) \xi^{-2} J \\ &\quad + \xi^{-1} \left(3 : JJ : + \frac{3(k+1)}{2} \partial J - (k + 3)T \right). \end{aligned}$$

Proof. Equality of every mode product gives the commutative square coefficient by coefficient. The identities $Y(\partial a, \xi) = \partial_\xi Y(a, \xi)$ and $[\partial_F, Y(a, \xi)] = \partial_\xi Y(a, \xi)$ prove compatibility with both connections. The vertex skew-symmetry and Jacobi identities give the chiral identities after taking principal parts and converting to right modules. The vacuum embedding is $g dz \mapsto g dz \otimes \mathbf{1}$. Its collision is $\text{pp}_\xi(\xi^{-1}Y(\mathbf{1}, \xi)a) = \xi^{-1}a$. Localization and tensor products over $\mathbb{C}$ preserve quasi-isomorphisms. The same holds for the direct sum $\xi^{-1}\mathbb{C}[\xi^{-1}]$. For a cochain map $\mu : P \rightarrow R$, use $\text{Cone}(\mu)^n = R^n \oplus P^{n+1}$ and $d(r, p) = (d_R r + \mu p, -d_P p)$. The cone's short exact sequence gives the cone quasi-isomorphism. The two displayed collisions follow directly from (1.5)–(1.6). The same coefficient argument preserves every finite iterated vertex product. $\square$

There is no strict unital vertex map $C_k \rightarrow W_k$ preserving cohomological degree. It would kill b_i, c_i and contradict $b_{i(0)}c_i = \mathbf{1}$. The inverse to ι is therefore an inverse in the category obtained by inverting quasi-isomorphisms. The polynomial family and its binary collision maps specify no comparison with the reflected Koszul dual. A completed chiral bar, a second differential, and descent through nodal curves require their own constructions.

1.2 AUGMENTATION, NORMAL PRODUCTS, AND THE SUBREGULAR POISSON QUOTIENT

The strict reduction map has the affine-ghost complex as its target. The vacuum product $J_{(1)}J$ prevents replacing that target by the bare affine algebra at level zero. It also restricts augmented bar constructions before any question of completion arises.

Use the universal polynomial family $W_R \subset C_R$ of Theorem 1.1.2, with $R = \mathbb{C}[k]$. The fields J, U, P, V retain their definitions and cohomological degree zero. At a complex level k , write W_k, C_k for their fibers. A vertex augmentation is a unital vertex homomorphism to $\mathbb{C}$ with zero translation and only its ordinary multiplication.

PROPOSITION 1.2.1 (*The affine target and the vacuum term*). There is no unital vertex homomorphism

$$W_o \longrightarrow V^o(\mathfrak{sI}_3).$$

More generally, W_k has a vertex augmentation if and only if $k = -3/2$. No pair W_k, W_{-k-6} admits vertex augmentations on both factors.

Proof. The universal affine algebra $V^o(\mathfrak{sI}_3)$ has an augmentation that kills every current. Its defining brackets permit this map: the affine central term is zero at level zero, and every Lie bracket maps to zero. Composing a proposed map from W_o with this augmentation gives a vertex augmentation of W_o . However, $J_{(1)}J = \mathbf{1}$ in W_o , while every positive product in $\mathbb{C}$ vanishes. This would give $0 = \mathbf{1}$.

At arbitrary k , the same argument applied to $J_{(1)}J = (2k+3)\mathbf{1}/3$ requires $k = -3/2$. At that level, all scalar terms in the universal presentation (1.5)–(1.6) vanish. Sending J, G^+, G^-, T to zero therefore defines a vertex augmentation by the presentation. Here $k+3 = 3/2$, so this presentation is defined. Finally,

$$\frac{2k+3}{3} + \frac{2(-k-6)+3}{3} = -2.$$

Both coefficients cannot vanish over $\mathbb{C}$. The same contradiction applies over every nonzero $\mathbb{C}$ -algebra. $\square$

The ghost target has no such augmentation because $b_{i(o)}c_i = \mathbf{1}$. Thus the explicit map $\mathcal{W}_o \hookrightarrow C_o$ is consistent with this obstruction. The proposition excludes a bare-affine inclusion, not the reduction map.

PROPOSITION 1.2.2 (*The first normal-product associator*). At every complex level, including $k = -3$, the polynomial fields satisfy

$$(: JJ :)U : - : J(: JU :) := 2 : (\partial J)U : \neq 0. \quad (1.8)$$

Consequently the ordinary tensor-bar coderivation defined only from the normal product does not square to zero. This failure occurs at tensor length three. There is no uncurved $\mathcal{A}_\infty$ structure with $m_1 = 0$ and $m_2(a, b) =: ab :$ on this state space.

Proof. For even states a, b , the vertex iterate identity gives

$$(: ab :)_{(-1)}c - a_{(-1)}b_{(-1)}c = \sum_{r \geq 0} (a_{(-r-2)}b_{(r)}c + b_{(-r-2)}a_{(r)}c).$$

Apply this with $a = b = J$ and $c = U$. The relations $J_{(0)}U = U$ and $J_{(r)}U = 0$ for $r \geq 1$ leave exactly $2J_{(-2)}U$. Translation identifies this state with $2 : (\partial J)U :$. Its leading PBW symbol is the nonzero monomial $2(\partial J)U$ in the polynomial jet algebra. Theorem 1.1.2 proves that it remains nonzero in every fiber.

Use the cohomological desuspension $sF = F[1]$, with $|sa| = |a| - 1$. On $T^c(sF) = \bigoplus_{n \geq 0} (sF)^{\otimes n}$, let b_2 be the coderivation with $b_2(sa \otimes sb) = (-1)^{|a|}s(: ab :)$. For these degree-zero fields, its square gives

$$b_2^2(sJ \otimes sJ \otimes sU) = 2s(: (\partial J)U :) \neq 0.$$

At $k = -3/2$, the same word lies in the augmentation ideal. Thus passing to that ideal does not remove this failure. In an uncurved $\mathcal{A}_\infty$ algebra with $m_1 = 0$, the arity-three identity is associativity of m_2 . Higher m_n cannot change that identity. $\square$

The proposition concerns this specified normal-product bar. A curved chiral bar has additional operations and configuration-space coefficients. Its definition must supply the terms that replace the failed identity (1.8). The obstruction does not exclude a separately constructed curved comparison at reflected levels.

1.2.1 The finite Poisson reduction

For a differential vertex superalgebra F , define

$$C_2(F) = \text{span}\{a_{(-2)}b : a, b \in F\}, \quad R_F = F/C_2(F).$$

The differential preserves $C_2(F)$. The quotient has an associative supercommutative product and a degree-zero super-Poisson bracket

$$\bar{a} \bar{b} = \overline{a_{(-1)}b}, \quad \{\bar{a}, \bar{b}\} = \overline{a_{(0)}b}.$$

Translation vanishes in this quotient. Quasi-commutativity, the iterate identity, and the vertex Jacobi identity prove these assertions. They are the C_2 construction in [3, Proposition 3.15, p. 21]. The quotient map retains these two induced operations. It is not a vertex homomorphism to a constant-field vertex algebra.

The affine and charged-fermion PBW bases identify

$$R_{C_R} = K_R = R[a, e, h, j, u, v, A, B] \otimes \Lambda(b_1, b_2, c_1, c_2). \quad (1.9)$$

Here the same symbols denote their quotient classes. The bracket on the affine variables is matrix commutation, while $\{b_i, c_j\} = \{c_j, b_i\} = \delta_{ij}$. The affine central coefficient and every derivative have disappeared. For the affine quotient this is [3, Eq. (3.19), p. 24]. For the charged ghosts it is the calculation in [3, Example 4.10, pp. 27–28]. Their tensor product gives (1.9). The same ordered-mode argument applies over R because all defining coefficients are polynomial in k .

Use ordinary supercommutative multiplication in K_R and set

$$\begin{aligned} H &= h + 2b_1c_1 + b_2c_2, & J &= j + b_2c_2, & U &= u - b_2c_1, & X &= v - b_1c_2, \\ p &= A + H^2/4 + UX, & v_o &= B - HX/2 - 3JX/2. \end{aligned} \quad (1.10)$$

Let $A_R = R[J, U, p, v_o]$, with zero differential. The inclusion $i : A_R \rightarrow K_R$ is given by these explicit polynomials.

THEOREM 1.2.3 (*Reduction after the C_2 quotient*). The natural map induced by $W_R \hookrightarrow C_R$ identifies R_{W_R} with A_R and gives a quasi-isomorphism

$$(R_{W_R}, \circ) = (A_R, \circ) \xrightarrow{i} (K_R, D)$$

of differential Poisson superalgebras over R . Its cohomology is concentrated in degree zero, and these assertions commute with every specialization of k . The Poisson bracket on A_R is

$$\begin{aligned} \{J, U\} &= U, & \{J, v_o\} &= -v_o, & \{J, p\} &= 0, \\ \{U, v_o\} &= p - \frac{2}{4}J^2, & \{p, U\} &= -\frac{3}{2}JU, & \{p, v_o\} &= \frac{3}{2}Jv_o. \end{aligned}$$

Proof. The differential on K_R is the differential of C_R with every derivative set to zero. Write $a' = a - 1$. The triangular polynomial change to coordinates

$$(a', e, b_1, b_2, H, X, c_1, c_2, J, U, p, v_o)$$

is invertible: first recover h, j, u, v from H, J, U, X and the ghosts, then recover A, B from (1.10). In these coordinates,

$$D = a' \partial_{b_1} + e \partial_{b_2} - 2c_1 \partial_H + c_2 \partial_X, \quad DJ = DU = Dp = Dv_o = 0.$$

Odd derivatives act from the left. The odd operator

$$s = b_1 \partial_{a'} + b_2 \partial_e - \frac{1}{2} H \partial_{c_1} + X \partial_{c_2}$$

satisfies $Ds + sD = N$, where N counts the first eight coordinates. Division by the positive integer eigenvalues of N contracts their positive-degree part. The projection $\pi : K_R \rightarrow A_R$ sets those eight coordinates to zero. It is an algebra chain map, $\pi i = 1$, and $Ds/N + sD/N = 1 - i\pi$ on the positive- N part. Thus i is a quasi-isomorphism of differential associative algebras.

Every class in R_{W_R} is a polynomial in the images of the four strong generators. Indeed, a derivative belongs to $C_2(W_R)$, and the quotient product is associative. The resulting surjection $A_R \rightarrow R_{W_R}$

composed with $R_{W_R} \rightarrow H^\circ(K_R)$ is the isomorphism just proved. It follows that the surjection is injective and that $R_{W_R} = A_R$. Functoriality of C_2 makes i a Poisson map. Taking the zero products of the polynomial fields, or substituting their displayed polynomials in the matrix and ghost brackets, gives the six formulas. No inverse of a polynomial in k occurs. The coordinate change and contraction therefore prove the same assertions in every fiber. $\square$

The projection π need not preserve the Poisson bracket. For example, $\pi h = 0$, $\pi a = 1$, and $\pi\{h, a\} = 2 \neq \{\pi h, \pi a\}$. This distinction matters for the bar construction below, which uses associative multiplication.

1.2.2 An actual associative bar comparison

Give A_R the augmentation $\epsilon_A(J) = \epsilon_A(U) = \epsilon_A(p) = \epsilon_A(v_0) = 0$. Give K_R the chain augmentation $\epsilon_K = \epsilon_A\pi$. In original coordinates it sends a to one and every other affine or ghost variable to zero. These are augmentations of differential associative algebras. They are not asserted to preserve the Poisson bracket or lift to vertex augmentations. Both i and π preserve them.

For an augmented differential associative algebra E , put $\bar{E} = \ker \epsilon_E$ and

$$B(E) = \bigoplus_{n \geq 0} (s\bar{E})^{\otimes_R n}, \quad |sa| = |a| - 1.$$

Use deconcatenation and the tensor Koszul signs. Its bar differential is the coderivation determined by

$$b_1(sa) = -s(Da), \quad b_2(sa \otimes sb) = (-1)^{|a|} s(ab).$$

THEOREM 1.2.4 (Direct-sum bar comparison). The maps $B(i)$ and $B(\pi)$ are explicit coalgebra chain maps and quasi-isomorphisms

$$B(A_R) \begin{matrix} \xrightarrow{B(i)} \\ \xleftarrow{B(\pi)} \end{matrix} B(K_R), \quad B(\pi)B(i) = 1.$$

On a word, they apply i or π separately to every letter. The construction commutes with every specialization of k .

Proof. The differential squares to zero by $D^2 = 0$, the Leibniz rule, and associativity. The signs displayed above give these three identities in arities one, two, and three, respectively. The maps preserve the unit, augmentation, multiplication, and differential, so their tensorwise extensions commute with b_1 , b_2 and deconcatenation.

The contraction in Theorem 1.2.3 preserves the augmentation kernels. All these complexes and maps are obtained from complexes of $\mathbb{C}$ -vector spaces by tensoring with R . It follows that every finite tensor power of $s\bar{i}$ is a quasi-isomorphism. Filter $B(i)$ by tensor length. The quotient at length n is that tensor-power map with its internal differential. Its cone is acyclic. For any cycle in the total cone, solve the boundary equation in its highest length quotient and subtract that boundary. The largest nonzero length decreases. Since every word sum is finite, this procedure terminates and proves that $B(i)$ is a quasi-isomorphism. The equation $B(\pi)B(i) = 1$ and the same argument prove the assertion for $B(\pi)$. The contraction works over every fiber, so specialization gives the same maps and proof. $\square$

This bar comparison applies to the full four-generator C_2 quotient of the subregular algebra. The quotient construction and Theorem 1.2.3 provide its comparison with the affine-ghost quotient. It

retains no first-product operation and kills every derivative and the associator (1.8). Indeed, $\overline{\partial J} = 0$ but $(\partial J)_{(1)} \overline{U} = -U \neq 0$. Thus the first product cannot descend to this quotient at any level. This prevents using the theorem as a proof about the unreduced vertex carrier or a completed chiral bar.

COROLLARY 1.2.5. Let $z_1 = J, z_2 = U, z_3 = p, z_4 = v_0$ in A_R . The cohomology of either bar in Theorem 1.2.4 is

$$H^{-n} \simeq \Lambda_R^n(R^4) \quad (0 \leq n \leq 4), \quad H^r = 0 \quad \text{otherwise.}$$

The class indexed by $i_1 < \cdots < i_n$ is represented in $B(A_R)$ by

$$\sum_{\sigma \in S_n} \text{sgn}(\sigma) [z_{i_{\sigma(1)}} | \cdots | z_{i_{\sigma(n)}}].$$

Applying $B(i)$ gives its explicit representative in $B(K_R)$.

Proof. The normalized free bar resolution has terms $A_R \otimes_R \tilde{A}_R^{\otimes n}$ in degree $-n$ and differential

$$\begin{aligned} d(a_0 | a_1 | \cdots | a_n) &= -a_0 a_1 | a_2 | \cdots | a_n \\ &\quad + \sum_{i=1}^{n-1} (-1)^{i-1} a_0 | \cdots | a_i a_{i+1} | \cdots | a_n. \end{aligned}$$

Its augmentation is ϵ_A . The operator $h(a_0 | a_1 | \cdots | a_n) = -1 | \tilde{a}_0 | a_1 | \cdots | a_n$, with $\tilde{a}_0 = a_0 - \epsilon_A(a_0)$, satisfies $dh + hd = 1 - \eta \epsilon_A$. Here $\eta : R \rightarrow A_R$ is the unit in degree zero, and the last term is zero in negative degrees. Thus this is a free resolution of R over A_R .

The polynomial Koszul complex $A_R \otimes \Lambda(\eta_1, \dots, \eta_4)$, with $|\eta_i| = -1$ and $d\eta_i = -z_i$, is another free resolution. The sequence $z_1, \dots, z_4$ is regular: each successive quotient is a polynomial ring in the remaining variables over R . Tensoring its four two-term resolutions proves exactness in negative degrees. The antisymmetrization in the statement, with initial coefficient 1, defines a chain map from this Koszul resolution to the bar resolution. Its internal product terms cancel in pairs by commutativity, while its first terms reproduce $d\eta_i = -z_i$. It lifts the identity of R . The usual inductive lifting of maps between projective resolutions gives a homotopy inverse. Tensoring with R over A_R therefore computes the bar cohomology by the zero-differential exterior algebra on the four η_i . Theorem 1.2.4 supplies the same result for $B(K_R)$. $\square$

THEOREM 1.2.6 (Induced-module bar comparison). Let M be a differential left A_R -module and put $M_K = K_R \otimes_{A_R} M$ using i . For $E = A_R, K_R$, form the direct-sum module bar

$$B(E; M_E) = \bigoplus_{n \geq 0} (s\tilde{E})^{\otimes n} \otimes_R M_E, \quad M_{A_R} = M.$$

Its differential consists of the internal differentials, the adjacent b_2 above, and the last-letter action $sa \otimes m \mapsto am$, with the tensor Koszul signs. The maps

$$[a_1 | \cdots | a_n] \otimes m \mapsto [i(a_1) | \cdots | i(a_n)] \otimes (1 \otimes m)$$

and the reverse map applying π to the letters and $x \otimes m \mapsto \pi(x)m$ to the module are chain quasi-isomorphisms. Their reverse composite on $B(A_R; M)$ is the identity. They commute with the natural bar comodule structures relative to $B(i)$ and $B(\pi)$.

Proof. The module differential squares to zero by the differential-module Leibniz identity and the associative action identity. For example, on two degree-zero letters its action terms are $(ab)m - a(bm) = 0$; on a homogeneous last letter the sign-free action anticommutes with the internal differential because $b_1(sa) = -s(Da)$. The displayed maps preserve all these operations.

In the coordinates of Theorem 1.2.3, $K_R = A_R \otimes_R Q_R$, where Q_R is the algebra on the eight contracted coordinates. The contraction s/N is A_R -linear. It therefore gives a contraction of $M_K \rightleftharpoons M$ for every differential M , with the tensor signs. The contraction of $s\tilde{K}_R \rightleftharpoons s\tilde{A}_R$ likewise gives homotopy equivalences on all finite tensor powers, including after tensoring with M . The associated map at every tensor length is consequently a quasi-isomorphism. The finite highest-length argument in Theorem 1.2.4 proves the assertion on the direct sums. This proof does not require boundedness of M or completion. $\square$

For $M = R$ with its augmentation action, $B(A_R; R) = B(A_R)$, so the theorem computes the cohomology of $B(K_R; K_R \otimes_{A_R} R)$ by the same exterior algebra. For a general affine vertex module there is no assertion here that its C_2 quotient is an induced module M_K . Constructing that comparison, and restoring the lost first products, are separate requirements.

1.2.3 Changing the central-charge parameter

Over $\mathbb{C}[k, (k+3)^{-1}]$, the central charge is $c = 25 - 6t - 24/t$, where $t = k+3$. Changing the base from the level to c is the finite algebra

$$S = \mathbb{C}[c, t] / (6t^2 + (c - 25)t + 24).$$

Its element t is invertible, with $t^{-1} = -(6t + c - 25)/24$. Eliminating c gives $S \simeq \mathbb{C}[t, t^{-1}]$. As a $\mathbb{C}[c]$ -module, S is free on $1, t$ because the relation is monic after division by 6. Its discriminant is $(c-1)(c-49)$. Thus the two branches are exchanged by $t \mapsto 4/t$ and meet at $c = 1, 49$. These equations define the coefficient change without a square root of c or an adic completion. Choosing a formal branch at a specified point is an additional base change.

1.2.4 Orbit dimensions and generator weights

PROPOSITION 1.2.7. For the Dynkin good grading of each nilpotent orbit in $\mathfrak{sl}_4$, the universal reduction at $k \neq -4$ has the following even strong generators.

partition	$\dim O$	$\dim \mathfrak{sl}_4^f$	generator weights
(1, 1, 1, 1)	0	15	1^{15}
(2, 1, 1)	6	9	$1^4, (3/2)^4, 2$
(2, 2)	8	7	$1^3, 2^4$
(3, 1)	10	5	$1, 2^3, 3$
(4)	12	3	$2, 3, 4$

An exponent records multiplicity. In particular, the subregular $\mathfrak{sl}_4$ algebra has five generators, with integral weights $1, 2, 2, 2, 3$.

Proof. Let L_m be the irreducible $\mathfrak{sl}_2$ -module of highest weight m and dimension $m+1$. A Jordan block of size d gives the summand L_{d-1} in the defining representation. The decompositions of its traceless endomorphisms, in the order of the table, are

$$15L_0, \quad 4L_0 \oplus 4L_1 \oplus L_2, \quad 3L_0 \oplus 4L_2, \quad L_0 \oplus 3L_2 \oplus L_4, \quad L_2 \oplus L_4 \oplus L_6.$$

They follow from $L_a \otimes L_b = \bigoplus_{r=0}^{\min(a,b)} L_{a+b-2r}$ and removal of the scalar endomorphism. In each L_m , the kernel of the lowering operator is one-dimensional and has grading degree $-m/2$. Kac–Wakimoto’s structure theorem therefore gives one even strong generator of weight $1 + m/2$ for each summand [1, Theorem 4.1, pp. 10–12]. Counting these summands gives the centralizer dimensions. Subtracting each from $\dim \mathfrak{sl}_4 = 15$ gives the orbit dimensions.

For the subregular row, this can also be checked directly. Take $f = E_{21} + E_{32}$ and $x = \text{diag}(1, 0, -1, 0)$. Its centralizer has basis

$$\text{diag}(1, 1, 1, -3), \quad f, \quad E_{34}, \quad E_{41}, \quad f^2,$$

with grading degrees $0, -1, -1, -1, -2$. □

COROLLARY 1.2.8. For $n \geq 2$ and noncritical levels $k, \ell \neq -n$, there is no conformal vertex isomorphism

$$\mathcal{W}^k(\mathfrak{sl}_n, f_{\text{prin}}) \simeq V^\ell(\mathfrak{sl}_n)$$

between the universal algebras with their standard conformal vectors.

Proof. The principal centralizer has basis $f, f^2, \dots, f^{n-1}$ with grading degrees $-1, -2, \dots, -(n-1)$. The same structure theorem gives strong generator weights $2, 3, \dots, n$. Its weight-one space is therefore zero. The universal affine algebra has weight-one space $\mathfrak{sl}_n$, of dimension $n^2 - 1$. A conformal isomorphism preserves these weight spaces. □

Partition transpose describes nilpotent orbit duality in type $\mathcal{A}$. The weight-space obstruction shows that this finite combinatorial operation does not, by itself, identify the vertex algebras attached to the two orbits.

PROPOSITION 1.2.9 (*A vacuum obstruction to generator truncation*). Let $\mathcal{W}_3$ have the normalization $W_{(3)}W = (c/3)\mathbf{1}$. If $c \neq 0$, no unital vertex homomorphism out of this algebra can send W to zero. In particular, at $c = -30$, deleting the weight-three generator cannot define a map to a Virasoro algebra.

Proof. Applying such a homomorphism to $W_{(3)}W = (c/3)\mathbf{1}$ would give $0 = (c/3)\mathbf{1}$ in its nonzero target. At $c = -30$, the right side is $-10\mathbf{1}$. □

BIBLIOGRAPHY

- [1] V. G. Kac and M. Wakimoto, *Quantum reduction and representation theory of superconformal algebras*, Adv. Math. **185** (2004), 400–458. <https://arxiv.org/abs/math-ph/0304011>. Locators refer to version 2.
- [2] T. Arakawa, *Rationality of Bershadsky–Polyakov vertex algebras*, Comm. Math. Phys. **323** (2013), 627–633. <https://arxiv.org/abs/1005.0185>. Locators refer to version 4.
- [3] T. Arakawa, *Introduction to W -algebras and their representation theory*, Perspectives in Lie Theory, Springer INdAM Series **19** (2017), 179–250. <https://arxiv.org/abs/1605.00138>. Locators refer to version 2.